URBAN VISION AI — SAFEBUILD

Plan roadwork. Keep communities connected.

Sustainable Futures Innovation Challenge (SFIC 2026) — Track A: Infrastructure & Access Planning

READINESS DECLARATION & PROTOTYPE NOTICE:
Working prototype tested using real Kakkanad geographic data and hypothetical construction scenarios; field validation and authority review pending.
SafeBuild is a functional decision-support prototype built to demonstrate how municipal roadwork restrictions can be modeled deterministically to assess community accessibility and compare eligible construction arrangements before ground-breaking. It is not an officially deployed municipal software or statutory clearance authority.

1. Executive Summary

Rapid urbanization and major transit infrastructure construction across Indian Tier-1 and Tier-2 cities frequently introduce uncoordinated road closures that sever community access to essential facilities—most critically healthcare receiving centers, public transport interchanges, schools, and employment centers. While civil engineering specifications rigorously define structural workzones, the spatial accessibility impacts on surrounding neighborhoods are rarely modeled before construction begins. The consequences include compounded detour burdens on vulnerable neighborhoods, unmanaged rat-running through narrow residential lanes, and total severance of non-motorized access.

Urban Vision AI — SafeBuild addresses this operational gap by providing an open, verifiable scenario planning studio that enables urban planners, public works agencies, and civic advocates to evaluate proposed road restrictions before implementation. Rather than relying on ungrounded generative guesses or computationally opaque black boxes, SafeBuild couples an authentic 19,582-node OpenStreetMap road network with deterministic 5-state routing algebra, multi-phase construction schedule modeling, and a guarded AI analyst governed by strict verification safeguards.

2. Strategic Alignment & Core Mission

SafeBuild directly advances SFIC Track A objectives by embedding community equity and accessibility into urban capital project planning. The project is structured around three foundational principles:

1. Spatial Equity First	Measures how road restrictions affect distinct residential clusters differently, identifying isolated neighborhoods and preventing one-size-fits-all diversion plans.
2. Mathematical Determinism	Eliminates simulation randomness and hallucinated timings. All metrics are computed as reproducible shortest-path graph calculations with complete referential integrity.
3. Truthful Prototype Gating	Enforces strict prototype review gates. If candidate mitigation options lack signed authority clearance or physical feasibility, SafeBuild strictly emits NO_VERIFIED_OPTION.

3. The Urban Challenge: Roadwork Severance in Indian Cities

In rapidly growing Indian metropolitan regions, linear infrastructure projects—such as elevated metro rail corridors, flyovers, stormwater drain overhauls, and utility trenching—frequently occupy the primary carriage-width of key arterial roads for 12 to 36 months. Standard construction practice relies on piecemeal, reactive traffic advisories issued only after barricades are erected. This paradigm causes four systemic planning failures:

Asymmetric Neighborhood Detour Penalties	While vehicles originating directly on the arterial may face modest detours, feeder settlements located behind the corridor face compounding access bottlenecks, turning short trips into circuitous journeys exceeding 20-30% additional travel distance.
Unregulated Residential Spillover	When major arterial reaches close without modeled alternative capacity, diverted passenger vehicles flood narrow collector and residential streets never designed for arterial traffic volumes, creating localized safety hazards and pavement degradation.
Pedestrian & Non-Motorized Vulnerability	Operational interventions like single-lane vehicular contraflow frequently consume existing footway width or force pedestrians into active construction gutters, directly compromising walkability.
Lack of Pre-Construction Arrangement Auditing	Civil contractors typically execute work sequences based on contractor equipment logistics rather than community accessibility exposure, missing opportunities to sequence reaches in ways that minimize cumulative community disruption.

4. Representative Case Study: Kakkanad Corridor, Kochi

To ground the prototype in genuine spatial geometry, SafeBuild models the **Palarivattom–Kakkanad Road Corridor** in Ernakulam District, Kerala, India. This corridor represents a critical arterial spine connecting residential suburbs, government administrative headquarters (Kakkanad Civil Station), and major employment centers (Infopark / SmartCity Kochi). The corridor is currently undergoing major public infrastructure planning, making it an ideal empirical testbed for evaluating simulated construction closures.

Study Attribute	Case Study Value	Topological Verification
Geographic Extent	Kakkanad, Kochi [76.330°E – 76.360°E, 10.010°N – 10.035°N]	Full 32.8 km² routing buffer bounding box
Base Road Network	19,582 nodes; 11,675 motor vehicle directed edges	OpenStreetMap contributors (ODbL 1.0), 0 dangling refs
Pedestrian Permeability	16,842 directed edges (footways, paths, steps)	Explicitly separated dual-graph routing topology
Evaluated Facility	Sunrise Hospital, Seaport-Airport Road	Verified active tertiary emergency healthcare anchor
Sampled Community Anchors	7 distinct residential clusters (Kakkanad West, Padamugal, DLF, etc.)	Curb-snapped centroid anchors representing settlement catchments
Simulated Restriction	Palarivattom–Kakkanad Rd Reach (Ways 197827530, 383253303, etc.)	Hypothetical civil pier foundation workzone closure

5. The SafeBuild Solution: Guided Four-Step Workflow

SafeBuild redesigns the planning interaction around one intuitive question: *'If this road is restricted during construction, how does community access change, and would another arrangement help?'* The application organizes complex network graph theory into four sequential steps:

Step 1: Define Work	Choose any road segment on the interactive map or select from pre-configured corridor phases. Specify directionality (bidirectional vs one-way), travel mode, and construction activity. All unverified closures are clearly tagged as hypothetical models.
Step 2: Check Access	Execute live Dijkstra shortest-path calculations against baseline network conditions. Instantly view modeled detours, newly unreachable origins, and visual before/after routing paths to critical facilities.
Step 3: Compare Arrangements	Audit candidate work arrangements (e.g. sequential phasing vs contraflow lane management vs accelerated concurrent closures) to understand trade-offs across duration, detour severity, and pedestrian mode conflict.
Step 4: Export Plan	Download RFC 4180 compliant CSV audit trails and tamper-evident PDF scenario briefs containing complete metadata, snapshot dates, and scientific disclaimers.

6. Formal 5-State Routing Algebra

To guarantee mathematical rigor and prevent confounding baseline network characteristics with construction impacts, SafeBuild defines a formal 5-state graph algebra. Every reported distance is strictly attributed to a specific network state:

State Notation	Formal Set Definition	Physical Meaning in Planning	Invariance / Safeguard
G0	G_base = (V, E_base)	Pre-construction baseline network	Baseline reference for all detour deltas
G0 + P	G_base ∪ E_proposal	Network with planned permanent capital additions	Identity Invariant: G0+P == G0 when P=identity
G0 + P + C	(G0 ∪ E_p) \ E_closure	Network under active construction restriction	Closure Monotonicity: dist(G0_C) >= dist(G0)
G0 + P + C + M	((G0 ∪ E_p) \ E_c) ∪ E_mitigation	Construction state with candidate mitigation (e.g. contraflow)	Re-weighted or added capacity; evaluated for feasibility

Attribution Isolation Principle:

Net construction detour delta is strictly defined as: Δ **= dist(G0_C) - dist(G0)**. Because baseline travel distance varies significantly across residential origins (e.g., DLF City is 2,060 m from Sunrise Hospital while Padamugal is 283 m), evaluating absolute detour distance alongside percentage increase provides an honest, transparent metric of impact.

7. Technical Architecture & Multi-Phase Engine

SafeBuild is architected as an offline-capable, high-performance web platform combining a FastAPI/NetworkX backend with a React/Next.js/MapLibre GL client. All spatial calculations execute deterministically on directed multigraphs without external API dependencies.

Geospatial Ingestion Engine	Parses raw OpenStreetMap XML files across tiled subboxes with 100% boundary overlap. Resolves shared nodes, eliminates duplicates, and builds referentially complete directed multigraphs (OSMParser).
Dual-Mode Directed Routing Engine	Maintains separate topological subgraphs for motor vehicles (excluding dedicated footpaths and service alleys) and pedestrians (incorporating stairs, living streets, and pedestrian paths) using NetworkX.
Multi-Phase Construction Engine	Models complex multi-phase construction schedules with predecessor dependencies. Features DFS-based circular dependency detection, calendar day unit normalization, and non-overlapping schedule exposure calculation.
Guarded Geospatial AI Analyst	Integrates Google Gemini API with deterministic server-side assertion filtering. If API credentials are unavailable or if prompt responses fail truthfulness filters, the engine falls back gracefully to deterministic analysis templates.

8. Multi-Phase Construction Schedule Specification

Real-world urban infrastructure works are never executed in a single monolithic closure. SafeBuild models construction as a sequence of discrete engineering activities with independent durations and road restrictions:

Phase ID	Engineering Activity	Duration	Dependencies	Restricted OSM Ways	Modeled Schedule Window
phase_1_utility	Underground Utility Relocation	21.0 days	None (Initial)	Way 197827530 (Eastbound)	Day 0.0 to 21.0
phase_2_piling	Metro Pier Foundation & Piling	45.0 days	Depends on Phase 1	Ways 197827530, 383253303 (Both)	Day 21.0 to 66.0
phase_3_girder	Viaduct Girder Launching	30.0 days	Depends on Phase 2	Ways 1011679137, 1011679138 (Both)	Day 66.0 to 96.0

Schedule Verification Note: The 96.0-day sequential duration represents the exact arithmetic sum of modeled activities (21 + 45 + 30 days). In accelerated scenarios (e.g. Arrangement C), an illustrative 35% time compression factor (0.65 multiplier -> 62.4 days) is explored to model parallel reach execution; this factor is explicitly labeled as a simulation assumption rather than an empirical CPM schedule.

9. Empirical Scenario Study Results: Access to Sunrise Hospital

The baseline analysis evaluates motor vehicle access from seven representative residential anchors across Kakkanad to the primary emergency entrance of Sunrise Hospital on Seaport-Airport Road. The hypothetical workzone restricts the Palarivattom–Kakkanad Road arterial reach (OSM Way IDs 197827530, 383253303, 1011679137, 1011679138):

Community Anchor (Origin)	Sampling Method	Baseline G0 (m)	Under Work G0_C (m)	Net Detour (m)	Detour %	Access Status
Kakkanad West / Civil Station	Road-side curb anchor	1,572.6	1,927.5	+354.9	+22.6%	Reachable via Collector
Civil Station Administrative Hub	Frontage curb node	1,842.1	2,052.4	+210.3	+11.4%	Reachable via Bypass
Padamugal Residential Anchor	Post-office frontage node	282.7	415.8	+133.1	+47.1%	Reachable via Local Lane
Kollamkudimugal Settlement	Junction centroid	1,140.5	1,140.5	0.0	0.0%	Unaffected (Alternate Corridor)
Canal Colony / Athani Settlement	Canal access road centroid	1,420.0	1,420.0	0.0	0.0%	Unaffected (Eastern Reach)
DLF City / New Town Heights	Arterial access centroid	2,060.4	2,060.4	0.0	0.0%	Unaffected (Seaport Corridor)
NGO Quarters Housing Colony	Colony arterial connector	1,650.2	1,650.2	0.0	0.0%	Unaffected (Northern Corridor)

10. Analytical Insights & Spatial Equity Findings

1. Asymmetric Severance Pattern: The closure creates pronounced spatial asymmetry. Western origins (Kakkanad West, Civil Station) and central residential pockets (Padamugal) bear the entire network detour penalty, while eastern and southern clusters (DLF City, Canal Colony) experience exactly 0.0 m change due to independent access via Seaport-Airport Road.

2. Percentage vs Absolute Disproportion: Padamugal experiences the highest relative route expansion (+47.1%), while Kakkanad West suffers the largest absolute distance penalty (+354.9 m). Evaluating both metrics simultaneously prevents misleading prioritization.

3. Zero Network Disconnection: Despite the closure of the primary arterial, 7 of 7 sampled origins retain connected routing paths to Sunrise Hospital via secondary collectors (e.g. Civil Lane Road and residential bypasses), demonstrating that the road network has topological redundancy.

4. Pedestrian Mode Divergence: Walking route distances are substantially shorter (1,240.2 m for Kakkanad West vs 1,572.8 m vehicle baseline) due to pedestrian permeability through narrow public footpaths. However, vehicular workzones directly threaten this permeability.

11. Bounded Work Arrangement Tradeoff Analysis

Rather than proposing unconstrained, arbitrary diversions, SafeBuild performs a bounded comparison over three operationally distinct construction sequencing arrangements, evaluating what improves, what worsens, and what remains unverified:

Arrangement A: Standard Sequential Phasing | *Sequential (96.0 days)*
Description: Executes utility relocation, piling, and girder launching one phase at a time.
Tradeoff Analysis:
• Improves: Restricts only one reach at any given time, preserving collector bypass capacity.
• Worsens: Extends total project schedule exposure to full 96.0 days.
Feasibility Status: PROVISIONAL_GEOMETRY_MODEL / UNVERIFIED_MODEL_SCENARIO | **Review Gate: Ineligible (Awaiting formal authority review)**

Arrangement B: Managed Contraflow & Signal Priority | *Sequential (96.0 days)*
Description: Preserves one managed unidirectional vehicle lane through the workzone with human flaggers.
Tradeoff Analysis:
• Improves: Mathematically recovers arterial connectivity, reducing worst detour by up to 180 m.
• Worsens: Compresses pedestrian walkway clearance and elevates worker exposure in the active workzone.
Feasibility Status: PROVISIONAL_GEOMETRY_MODEL / UNVERIFIED_MODEL_SCENARIO | **Review Gate: Ineligible (Awaiting safety review & authority clearance)**

Arrangement C: Accelerated Concurrent Work | *Concurrent (62.4 days modeled)*
Description: Simultaneously closes multiple corridor reaches to compress modeled project duration by ~35%.
Tradeoff Analysis:
• Improves: Shortens total calendar duration exposure from 96.0 to 62.4 days.
• Worsens: Simultaneous corridor closures concentrate detour traffic onto collector streets, compounding neighborhood route congestion.
Feasibility Status: HIGH_NETWORK_DETOUR_CONCENTRATION / EXCLUDED_BY_MODEL_CONSTRAINTS | **Review Gate: Excluded (Fails scenario accessibility thresholds)**

12. The Prototype Review Gate: Strict NO_VERIFIED_OPTION

SYSTEM DECISION GATE OUTPUT: NO_VERIFIED_OPTION
Mandatory Ethical Safeguard: SafeBuild enforces a strict decision safeguard: no candidate routing alteration or diversion " may be presented to decision-makers as formal, implementable advice without documented physical site clearance and verified authority permits. " While Arrangement B demonstrates mathematical detour reduction, it introduces pedestrian mode conflicts and lacks verified field traffic management plans. " Arrangement C is excluded due to severe compounded detour concentration. Therefore, the optimizer safely and strictly outputs " NO_VERIFIED_OPTION, protecting against false algorithmic authority.

13. Cryptographic Data Provenance & Test Verification

To ensure end-to-end reproducibility, all source geographic data, test suites, and analytical outputs are backed by cryptographic checksums and automated verification suites:

OSM Tile Artifact	Spatial Coverage Area	SHA-256 Checksum
subbox_0.osm	Northwest Kakkanad (Vazhakkala / Palarivattom border)	50ebc99d4db060d844c2cf4d820384d250fb3c4cec774278e5f4a37efb98c88e
subbox_1.osm	North-central corridor (Padamugal arterial reach)	3a62f672ca598aa5c77206b598910e974eeb353bf472cf14bb637ecd2ab512e2
subbox_2.osm	Northeast sector (Kollamkudimugal residential)	9c4cc29acb749e2a81c42b9f87ed30aef31552a1cd17f5bbb878284f830ec6b6
subbox_3.osm	Southwest sector (Civil Station Environs / Seaport)	dc0799244624b702671612d18f9e22a4a9e7aa427acb9db36f54025177fcb6e5
subbox_4.osm	South-central corridor (Collectorate Road junction)	014a8255424f3b470ec8638cfd62d8f85eeefd59c239890f69d4055e5b82431e
subbox_5.osm	Southeast sector (Infopark Expressway / DLF City)	cb8cab0c22dae818d2816b04ad121e0b826aa340ae961ed7a0e8fcb7035f298f

14. Automated Test Suite Classification (17 Tests)

The test suite spans three test modules (`test_analytical_core.py`, `test_validation_protocol.py`, `test_multi_phase.py`) and verifies all 17 critical system invariants:

Test Category	Count	Verified Invariants & Test Scope	Status
Validation	6	Schema validation, zero missing highway refs, unknown OSM way guard, duration guards, prototype review gate	6 / 6 PASSED
Correctness	8	Directed oneway traversal, grade separation without junction, closure monotonicity, proposal algebra identity, circular dependency detection, schedule exposure sum, multi-phase execution	8 / 8 PASSED
Geographic	2	CRS coordinate bounds [76.330-76.360°E, 10.010-10.035°N], UTM Zone 43N metric projection consistency	2 / 2 PASSED
Guarded AI	1	Server-side assertion filter intercepting ungrounded causal claims and fabricated approvals	1 / 1 PASSED
Total Suite	17	Complete automated unit and invariant test suite across all engines	17 / 17 PASSED

15. Independent Reference Implementation Audit

To confirm accuracy beyond production self-testing, an independent verification script (`scripts/independent_verification.py`) was executed using a separate XML parser and standalone NetworkX graph. The results cross-verified against production:

Verification Metric	Independent Reference	Production Engine	Discrepancy	Tolerance & Outcome
Baseline G0 Distance	1,572.61 m	1,572.60 m	0.0055 m (5.5 mm)	Tol: 0.1 m -> EXACT AGREEMENT
Restricted G0_C Distance	1,927.50 m	1,927.50 m	0.0043 m (4.3 mm)	Tol: 0.1 m -> EXACT AGREEMENT
Net Detour Delta (Δ)	+354.89 m	+354.90 m	0.0098 m (9.8 mm)	Tol: 0.1 m -> EXACT AGREEMENT
Sequential Schedule	96.0 days	96.0 days	0.0 days	Exact arithmetic match
Concurrent Schedule (0.65x)	62.4 days	62.4 days	0.0 days	Exact match (illustrative assumption)

16. Proposed Implementation Budget & Operating Economics

Important Notice: The figures below represent a **proposed implementation budget** for a potential 3-month collaborative development and field-study cycle with local stakeholders. They do **not** represent past expenditures incurred.

Budget Component	Proposed Allocation (INR)	Scope & Key Deliverables
Geospatial Data Engineering	Rs. 60,000 – Rs. 1,10,000	High-resolution curb and sidewalk survey, micro-turning relation tagging, OSM reconciliation
Platform Development & Modeling	Rs. 1,10,000 – Rs. 2,00,000	Dynamic traffic diversion integration, multi-facility batch routing, scenario editor enhancements
Field Validation & Ground Survey	Rs. 60,000 – Rs. 1,15,000	Physical site surveys of proposed detours, pedestrian conflict cross-section measurement
Stakeholder Review & Workshops	Rs. 40,000 – Rs. 80,000	Workshops with local planning bodies, community consultations, feedback synthesis
Total Proposed 3-Month Cycle	Rs. 2,70,000 – Rs. 5,05,000	Complete 3-month research development and field scenario evaluation package

Ongoing Monthly Operating Model:

Following initial deployment, ongoing monthly operations are projected at **Rs. 8,000 – Rs. 25,000 / month**, covering cloud hosting (VPS/container runtime), periodic OSM network refreshes, server maintenance, and guarded AI API token usage. This lean operating cost ensures long-term civic sustainability.

17. Methodological Limitations & Ethical Safeguards

Static Shortest Path vs Dynamic Traffic	SafeBuild computes static shortest path distances. It does not perform dynamic traffic assignment (DTA), queue simulation, or live traffic delay estimation. Detour distance is a structural proxy for accessibility, not observed travel time.
No Fabricated Emergency Timing	The system never converts route distance into synthetic ambulance travel seconds or survival probability claims without empirical vehicle-telemetry datasets.
No Demographic Extrapolations	Impacts are reported across sampled geographic cluster centroids, not synthesized population counts or demographic voting statistics.
Planar Coastal Topography	Elevation and road slope impedances are omitted, which is appropriate for the coastal plains of Kochi but would require enhancement for hilly terrain.
Authority & Regulatory Boundary	SafeBuild outputs are decision-support scenario models. Real-world implementation strictly requires statutory review, formal traffic management plans, and inter-agency approval.

Conclusion: SafeBuild demonstrates that urban roadwork planning can move beyond uncoordinated closures and ungrounded claims. By pairing authentic geospatial data with transparent graph algebra, multi-phase scheduling, and ethical prototype safeguards, SafeBuild enables cities to plan construction while keeping communities connected.